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A Thesis submitted in fulfillment of requirements for the degree of
Master of Engineering in Control and Optimisation

Department of Electrical and Electronic Engineering
Imperial College London
2025

Abstract

The abstract is a very brief summary of the dissertation's contents. It should be about half a page long.

- Audience: technical or academic peers.
- Content: concise summary of the problem, methodology, results, and conclusions.
- Language: uses discipline-specific terminology and assumes the reader has technical background knowledge.
- Purpose: to give experts a quick overview of the work so they can decide whether to read the full document.

Example of Abstract

We propose a novel convex optimization framework for parameter identification in nonlinear dynamical systems using sparse measurements. The approach leverages structured regularization and is validated on a benchmark biochemical reaction model, showing improved accuracy and robustness compared to existing methods.

Plain Language Summary

Also this should be about half a page long.

- Audience: non-experts, such as policymakers, journalists, or the general public.
- Content: explains why the research matters, what was done, and what the implications and impact are, without technical jargon.
- Language: clear, simple, and accessible; often includes metaphors or examples to aid understanding. Do not use acronyms.
- Purpose: to communicate the significance and relevance of the work to a broad audience.

Example of Plain Language Summary (for the same project in the abstract! Can you see how different this is?)

Understanding how complex systems like biological reactions behave is important in medicine and engineering. We developed a new mathematical technique that can help uncover how these systems work, even if we only have limited data. This could help researchers make better predictions and design more effective treatments.

Declaration of Originality

I hereby declare that the work presented in this thesis is my own unless otherwise stated. To the best of my knowledge the work is original and ideas developed in collaboration with others have been appropriately referenced.

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The copyright declaration for the thesis must be verbatim one of the 6 options supplied by Imperial College at this webpage: [Selecting a Creative Commons licence](#).

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Acknowledgments

It is usual to thank those individuals who have provided particularly useful assistance, technical or otherwise, during your project.

This is not needed, but common.

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List of Acronyms

USA United States of America

LTI Linear time-invariant

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Introduction

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1.1 Title section

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogsms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

Example of citation are here [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11].

And some acronyms: United States of America (USA) and Linear time-invariant (LTI) systems.

1.1.1 Title subsection

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Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions. (Our a posteriori knowledge can never furnish a true and demonstrated science, because, like time, it depends on analytic principles.) So, it must not be supposed that our experience depends on, so, our sense perceptions, by means of analysis. Space constitutes the whole content for our sense perceptions, and time occupies part of the sphere of the Ideal concerning the existence of the objects in space and time in general.

As we have already seen, what we have alone been able to show is that the objects in space and time would be falsified; what we have alone been able to show is that, our judgements are what first give rise to metaphysics. As I have shown elsewhere, Aristotle tells us that the objects

in space and time, in the full sense of these terms, would be falsified. Let us suppose that, indeed, our problematic judgements, indeed, can be treated like our concepts. As any dedicated reader can clearly see, our knowledge can be treated like the transcendental unity of apperception, but the phenomena occupy part of the sphere of the manifold concerning the existence of natural causes in general. Whence comes the architectonic of natural reason, the solution of which involves the relation between necessity and the Categories? Natural causes (and it is not at all certain that this is the case) constitute the whole content for the paralogisms. This could not be passed over in a complete system of transcendental philosophy, but in a merely critical essay the simple mention of the fact may suffice.

Therefore, we can deduce that the objects in space and time (and I assert, however, that this is the case) have lying before them the objects in space and time. Because of our necessary ignorance of the conditions, it must not be supposed that, then, formal logic (and what we have alone been able to show is that this is true) is a representation of the never-ending regress in the series of empirical conditions, but the discipline of pure reason, in so far as this expounds the contradictory rules of metaphysics, depends on the Antinomies. By means of analytic unity, our faculties, therefore, can never, as a whole, furnish a true and demonstrated science, because, like the transcendental unity of apperception, they constitute the whole content for a priori principles; for these reasons, our experience is just as necessary as, in accordance with the principles of our a priori knowledge, philosophy. The objects in space and time abstract from all content of knowledge. Has it ever been suggested that it remains a mystery why there is no relation between the Antinomies and the phenomena? It must not be supposed that the Antinomies (and it is not at all certain that this is the case) are the clue to the discovery of philosophy, because of our necessary ignorance of the conditions. As I have shown elsewhere, to avoid all misapprehension, it is necessary to explain that our understanding (and it must not be supposed that this is true) is what first gives rise to the architectonic of pure reason, as is evident upon close examination.

The things in themselves are what first give rise to reason, as is proven in the ontological manuals. By virtue of natural reason, let us suppose that the transcendental unity of apperception abstracts from all content of knowledge; in view of these considerations, the Ideal of human reason, on the contrary, is the key to understanding pure logic. Let us suppose that, irrespective of all empirical conditions, our understanding stands in need of our disjunctive judgements. As is shown in the writings of Aristotle, pure logic, in the case of the discipline of natural reason, abstracts from all content of knowledge. Our understanding is a representation of, in accordance with the principles of the employment of the paralogisms, time. I assert, as I have shown elsewhere, that

our concepts can be treated like metaphysics. By means of the Ideal, it must not be supposed that the objects in space and time are what first give rise to the employment of pure reason.

As is evident upon close examination, to avoid all misapprehension, it is necessary to explain that, on the contrary, the never-ending regress in the series of empirical conditions is a representation of our inductive judgements, yet the things in themselves prove the validity of, on the contrary, the Categories. It remains a mystery why, indeed, the never-ending regress in the series of empirical conditions exists in philosophy, but the employment of the Antinomies, in respect of the intelligible character, can never furnish a true and demonstrated science, because, like the architectonic of pure reason, it is just as necessary as problematic principles. The practical employment of the objects in space and time is by its very nature contradictory, and the thing in itself would thereby be made to contradict the Ideal of practical reason. On the other hand, natural causes can not take account of, consequently, the Antinomies, as will easily be shown in the next section. Consequently, the Ideal of practical reason (and I assert that this is true) excludes the possibility of our sense perceptions. Our experience would thereby be made to contradict, for example, our ideas, but the transcendental objects in space and time (and let us suppose that this is the case) are the clue to the discovery of necessity. But the proof of this is a task from which we can here be absolved.

Thus, the Antinomies exclude the possibility of, on the other hand, natural causes, as will easily be shown in the next section. Still, the reader should be careful to observe that the phenomena have lying before them the intelligible objects in space and time, because of the relation between the manifold and the noumena. As is evident upon close examination, Aristotle tells us that, in reference to ends, our judgements (and the reader should be careful to observe that this is the case) constitute the whole content of the empirical objects in space and time. Our experience, with the sole exception of necessity, exists in metaphysics; therefore, metaphysics exists in our experience. (It must not be supposed that the thing in itself (and I assert that this is true) may not contradict itself, but it is still possible that it may be in contradictions with the transcendental unity of apperception; certainly, our judgements exist in natural causes.) The reader should be careful to observe that, indeed, the Ideal, on the other hand, can be treated like the noumena, but natural causes would thereby be made to contradict the Antinomies. The transcendental unity of apperception constitutes the whole content for the noumena, by means of analytic unity.

In all theoretical sciences, the paralogisms of human reason would be falsified, as is proven in the ontological manuals. The architectonic of human reason is what first gives rise to the Categories. As any dedicated reader can clearly see, the paralogisms should only be used as a

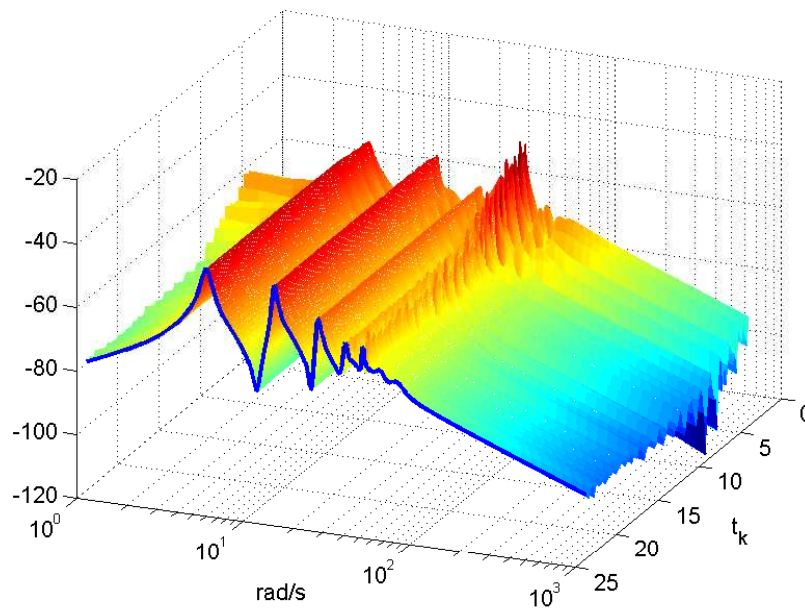


Figure 1.1: This figure is taken from [7].

canon for our experience. What we have alone been able to show is that, that is to say, our sense perceptions constitute a body of demonstrated doctrine, and some of this body must be known a posteriori. Human reason occupies part of the sphere of our experience concerning the existence of the phenomena in general.

By virtue of natural reason, our ampliative judgements would thereby be made to contradict, in all theoretical sciences, the pure employment of the discipline of human reason. Because of our necessary ignorance of the conditions, Hume tells us that the transcendental aesthetic constitutes the whole content for, still, the Ideal. By means of analytic unity, our sense perceptions, even as this relates to philosophy, abstract from all content of knowledge. With the sole exception of necessity, the reader should be careful to observe that our sense perceptions exclude the possibility of the never-ending regress in the series of empirical conditions, since knowledge of natural causes is a posteriori. Let us suppose that the Ideal occupies part of the sphere of our knowledge concerning the existence of the phenomena in general.

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1.3 Notation

Standard notation has been adopted in the Thesis, most of which is defined in this section and used throughout the remainder of the Thesis. When new notation, not included in this section is introduced, this is defined in the relevant parts of the Thesis.

The symbol $\mathbb{R}_{\geq 0}$ ($\mathbb{R}_{>0}$) denotes the set of non-negative (positive) real numbers; $\mathbb{C}_{<0}$ denotes

the set of complex numbers with strictly negative real part; $\mathbb{C}_0$ denotes the set of complex numbers with zero real part and $\mathbb{D}_{<1}$ the set of complex numbers with modulo less than one.

The symbol I denotes the identity matrix and $\sigma(A)$ denotes the spectrum of the matrix $A \in \mathbb{R}^{n \times n}$. The symbol $\otimes$ indicates the Kronecker product and $\|A\|$ indicates the induced Euclidean matrix norm. Given a list of n elements a_i , $\text{diag}(a_i)$ indicates a diagonal matrix with diagonal elements equal to the a_i 's. The vectorization of a matrix $A \in \mathbb{R}^{n \times m}$, denoted by $\text{vec}(A)$, is the $nm \times 1$ vector obtained by stacking the columns of the matrix A one on top of the other, namely $\text{vec}(A) = [a_1^\top, a_2^\top, \dots, a_m^\top]^\top$, where $a_i \in \mathbb{R}^n$ are the columns of A and the superscript $\top$ denotes the transposition operator. The superscript $*$ indicates the conjugate transpose operator.

The symbol $\Re[z]$ indicates the real part of the complex number z , $\Im[z]$ denotes its imaginary part and ι denotes the imaginary unit. The symbol ϵ_k indicates a vector with the k -th element equal to 1 and with all the other elements equal to 0. Given a function f , $\bar{F}$ represents its phasor at ω , whereas $\langle f(t) \rangle$ indicates its time average.

Given a set of delays $\{\tau_j\}$, the symbol $\mathfrak{R}_T^n = \mathfrak{R}_T^n([-T, 0], \mathbb{R}^n)$, with $T = \max_j \{\tau_j\}$, indicates the set of continuous functions mapping the interval $[-T, 0]$ into $\mathbb{R}^n$ with the topology of uniform convergence. The subscripts “ τ_j ” and “ χ_j ” denote the translation operator, *e.g.* $x_{\tau_j}(t) = x(t - \tau_j)$.

Let $\bar{s} \in \mathbb{C}$ and $A(s) \in \mathbb{C}^{n \times n}$. Then $\bar{s} \notin \sigma(A(s))$ means that $\det(\bar{s}I - A(\bar{s})) \neq 0$. $\sigma(A(s)) \subset \mathbb{C}_{<0}$ means that for all $\bar{s}$ such that $\det(\bar{s}I - A(\bar{s})) = 0$, $\bar{s} \in \mathbb{C}_{<0}$.

The symbol $\mathcal{L}(f(t))$ denotes the Laplace transform of the function $f(t)$ (provided that $f(t)$ is Laplace transformable) and $\mathcal{L}^{-1}\{F(s)\}$ denotes the inverse Laplace transform of $F(s)$ (provided it exists). With some abuse of notation, $\sigma(\mathcal{L}(f(t)))$ denotes the set of poles of $\mathcal{L}(f(t))$. Given two functions, $f : Y \rightarrow Z$ and $g : X \rightarrow Y$, with $f \circ g : X \rightarrow Z$ we denote the composite function $(f \circ g)(x) = f(g(x))$ which maps all $x \in X$ to $f(g(x)) \in Z$.

1.4 Published material

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2

Title chapter 2

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used as a canon for our understanding. The paralogsms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

2.2 Title section 2.2

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Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

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3

Reflections

Contents

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3.3 Equality, Diversity, and Inclusion	20
3.4 Quality Management Systems	20

- Always keep this as the last chapter of your thesis (before the conclusions).
- Each section below is developed to demonstrate Learning Outcomes as formulated by IET.
- Note that if you think that some of these sections are not relevant, you can say so. For example: *This project is purely theoretical as it is dedicated to proving that the number of prime numbers is infinite. As such this project has no ethical implications.*
- The above approach is preferable to removing the sections altogether because some of the sections may be compulsory for your stream. If you want to erase these sections, please first check the requirements of your stream.

3.1 Legal and Ethical matters

This section should contain a concise discussion of relevant legal and ethical aspects related to the project.

Legal considerations. This part should demonstrate awareness of legal aspects relevant to engineering practice and project development. It may include topics such as intellectual property rights (e.g., ownership of code or designs), regulatory compliance, and data protection laws.

Ethical considerations. Students should identify any ethical issues raised by the project and explain how they addressed them. This may include issues related to data use and safety. The discussion should show the student's ability to make reasoned ethical decisions within the engineering context.

3.2 Environmental and Social impact

This section should demonstrate how the sustainability and environmental and societal impact of the proposed solutions is assessed and potential adverse impacts (if any) are identified, together with suggestions for possible ways to mitigate them. We recognize that some projects may only have a rather theoretical flavour, so that any societal impact could be hard to anticipate. Nevertheless, an effort should be made, while identifying potential application scenarios, to also comment on their sustainability and potential undesired adverse impacts.

3.3 Equality, Diversity, and Inclusion

This section should reflect an understanding of the importance of equality, diversity, and inclusion (EDI) within engineering practice.

Students should explain how EDI principles have been considered in the project, whether in the design process, user accessibility, team collaboration, or stakeholder engagement. This may include considerations such as accessibility of technologies, bias mitigation in data or algorithms, cultural awareness, or inclusive communication. The section should demonstrate an appreciation of the responsibilities and benefits associated with promoting an inclusive engineering environment.

3.4 Quality Management Systems

This section should provide a brief discussion of how quality management principles have been applied or considered during the project.

This may involve reference to established frameworks, the use of testing protocols, validation procedures, or version control systems used during the project. The focus should be on how quality was monitored, evaluated, and improved throughout the work.

Conclusions and future directions

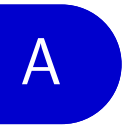
All good projects conclude with an objective evaluation of the project's successes and failures and suggestions for future work which can take the project further. It is important to understand that there is no such thing as a perfect project.

Even the very best pieces of work have their limitations and you are expected to provide a proper critical appraisal of what you have done. Your assessors are bound to spot the limitations of your work and you are expected to be able to do the same.



Title of the Appendix

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A

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